

The Third Law of Thermodynamics Nobody knows what entropy really is, so in any discussion you will always
 2 possible picture arrangements There are many possible ways of sticking pictures; I
 Large system has more entropy than the small system You can also increase the total

Now imagine you are sticking pictures on your wall A , nearby is my wall B . I ask if you could decorate my wall too, so our
 The total entropy of wall $A + B$ system When you start moving pictures to my wall

Inductively, we imagine the universe as a giant closed system, with many particles inside; and they are always exchanging "

An ice cube is more "organized" than a cup of water, and vice versa. In the middle of winter, water is frozen. Does this ph
 Water is frozen Surprisingly, the answer is no. The water is cooled down since it co

But:

Entropy is closely related to energy, but they are not the same, and we do not have any law of conservation of entropy; so w

So the total entropy of our system (a cup of water and the outside environment) increases, because it gains more than losses

And obviously, also by the law:

The temperature itself is not so important than the **difference**. If our system only consists objects at same temperature, n

In this section, we develop the connection among three physical quantities: (**heat, energy, entropy**). Let's begin with a ve

Plot our data into a simple graph: [H]
 A simple graph of the entropy Looking at table and graph, we find out some interest

Again, the total entropy is increasing! Can I guess the slope of a tangent line at point $q_A = 20$ is **steeper** than point $q_B =$

Substituting our values yields:

Here is a question, why is ΔS_B negative, but it still has positive slope? The answer is pretty simple, our graph perspective

According the zeroth law of thermodynamics, the temperature of solid A and B are equal if and only if they are in equilibri

A subtle observation is that a decrease in q_A corresponds to an increase in q_B :

This equation makes perfect sense to our intuition, look at previous calculation: the slope of S_A graph is much steeper than

To ensure the second law of thermodynamics, as you can see here, the entropy of solid A increases **faster** than the rate of e

Apparently, we can see that: